

Uni3R: Unified 3D Reconstruction and Semantic Understanding via Generalizable Gaussian Splatting from Unposed Multi-View Images

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https://horizonrobotics.github.io/robot_lab/uni3R/

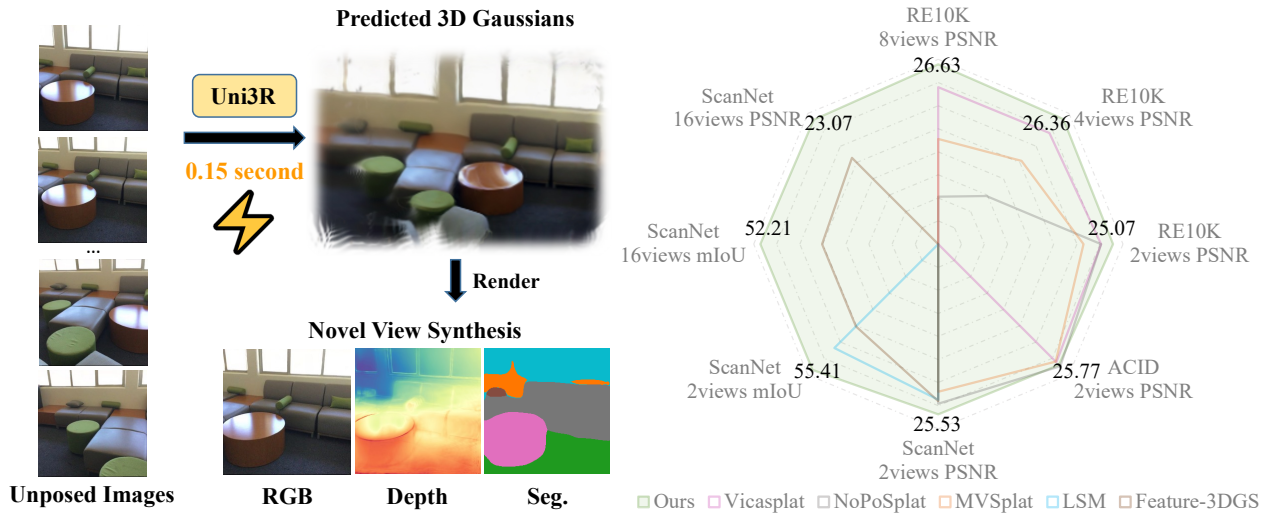


Figure 1. Uni3R takes unposed arbitrary multi-view images as input and produces a unified 3D Gaussian scene representation, enabling state-of-the-art performance in view synthesis, semantic segmentation, and depth estimation within a single forward pass.

Abstract

Reconstructing and semantically interpreting 3D scenes from sparse 2D views remains a fundamental challenge in computer vision. Conventional methods often decouple semantic understanding from reconstruction or necessitate costly per-scene optimization, thereby restricting their scalability and generalizability. In this paper, we introduce **Uni3R**, a novel feed-forward framework that jointly reconstructs a unified 3D scene representation enriched with open-vocabulary semantics, directly from unposed multi-view images. Our approach leverages a Cross-View Transformer to robustly integrate information across arbitrary

multi-view inputs, which then regresses a set of 3D Gaussian primitives endowed with semantic feature fields. This unified representation facilitates high-fidelity novel view synthesis, open-vocabulary 3D semantic segmentation, and depth prediction—all within a single, feed-forward pass. Extensive experiments demonstrate that Uni3R sets a new state of the art across multiple benchmarks, including in-domain datasets such as RE10K and ScanNet, as well as the out-of-domain dataset Mip-NeRF360. This work represents a new paradigm toward generalizable and unified 3D scene reconstruction and understanding.

1. Introduction

The ability to perceive and interpret the 3D world from sparse images is a cornerstone of computer vision, holding profound implications for robotics, autonomous driving, and augmented reality. While significant progress has been

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made in 3D reconstruction, led by photorealistic methods such as Neural Radiance Fields (NeRF) [24], 3D Gaussian Splatting (3DGS) [14], their reliance on time-consuming, per-scene optimization critically limits their generalizability to novel scenes. In response, a prominent class of generalizable 3D reconstruction methods [3, 5, 23, 39, 42] has emerged, which learn geometric priors across diverse scenes to perform feed-forward 3D reconstruction in a feed-forward manner.

While promising, these methods typically focus exclusively on geometry and appearance, overlooking the semantic richness crucial for holistic scene understanding. Recent efforts, including LangSplat [28] and Feature-3DGS [45], have incorporated semantic fields into 3D Gaussian Splatting, yet remain constrained by scene-specific optimization and lack scalability in real-world, zero-shot applications. More recently, approaches such as LSM [8] and UniForward [34] have aimed to unify semantic and radiance fields to jointly infer geometry, appearance, and semantics. However, these methods are built upon DUST3R [37], which is inherently designed for two-view inputs. Consequently, extending them to multi-view scenarios requires expensive pairwise feature matching across views, compromising efficiency and leading to inconsistent reconstructions due to the absence of global 3D context.

To address these limitations, we propose Uni3R, a novel, generalizable framework that synthesizes a unified 3D representation from arbitrary multi-view images for both high-fidelity rendering and dense, open-vocabulary semantic understanding. Leveraging a Cross-View Transformer effectively fuses information across views and produces globally consistent representations, Uni3R predicts unified 3D Gaussian primitives enriched with open-vocabulary semantic features. These Gaussian representations can be seamlessly rendered in real-time to synthesize novel views, supervised solely with source images and bypassing the need for per-scene optimization. Simultaneously, the embedded semantic features enable zero-shot 3D semantic segmentation by querying the scene with arbitrary text prompts.

To further enhance both geometric fidelity and training stability, we introduce a point-map-guided geometric loss that serves two key purposes. First, it enforces structural consistency and improves geometric accuracy, as evidenced by lower depth errors (e.g., AbsRel). Second, it stabilizes training by preventing the model from getting trapped in local minima when predicting the freedom 3D point distribution. Specifically, we employ a frozen VGGT [36] to generate dense point maps with associated confidence scores, which act as soft geometric priors to guide the spatial distribution of the 3D Gaussians.

Our contributions are summarized as follows:

- We introduce Uni3R, a novel feed-forward architecture that unifies 3D reconstruction and semantic understand-

ing. It predicts a set of Gaussian primitives with jointly integrated geometry, appearance, and open-vocabulary semantics in a single pass, eliminating the need for per-scene optimization.

- We demonstrate that a powerful geometry foundation model can be effectively extended beyond geometric estimation to support both photometric reconstruction and 3D scene understanding. Its cross-frame attention mechanism enables robust feature fusion to produce globally consistent scene representations from an arbitrary number of input views, while its predicted point maps provide potent geometric guidance.
- Uni3R achieves state-of-the-art performance across multiple tasks, including novel view synthesis, open-vocabulary 3D semantic segmentation, and depth prediction on the challenging RE10K [46] and ScanNet [6] datasets, underscoring its superior generalization and versatility.

2. Related Work

2.1. Differentiable Neural Representations

Traditional 3D reconstruction methods, such as Structure-from-Motion (SfM) [7] and Multi-View Stereo (MVS) [2] decompose the process into sequential steps, including feature matching, camera pose estimation, and geometric reconstruction. While effective, these multi-stage processes can be fragile and prone to error accumulation. The advent of Neural Radiance Fields (NeRF) [24] revolutionized novel view synthesis by introducing an end-to-end, differentiable approach that models a scene as a continuous function mapping 5D coordinates to color and volumetric density. More recently, 3D Gaussian Splatting (3DGS) [9, 14, 16, 33] has emerged as a compelling alternative, representing scenes explicitly with a set of 3D Gaussian primitives. Leveraging a highly efficient differentiable rasterizer, 3DGS supports real-time rendering speeds while maintaining exceptional rendering quality. However, canonical 3DGS relies on point clouds from SfM for initialization and rectified camera poses. Our work builds upon the 3DGS formulation but removes reliance on external tools like COLMAP [7]. By predicting Gaussians in an end-to-end pose-free manner, we enable scalable 3D reconstruction and scene understanding.

2.2. Feed-forward 3D Reconstruction

The substantial computational cost of per-scene optimization has motivated the development of 3D feed-forward models. PixelNeRF [42] and MVSNerf [4] learn scene priors across a large number of training scenes, enabling them to predict radiance fields for novel scenes from only a few input views. Following the success of 3DGS, pixelSplat [3], MVSplat [5], Generative Densification [25],

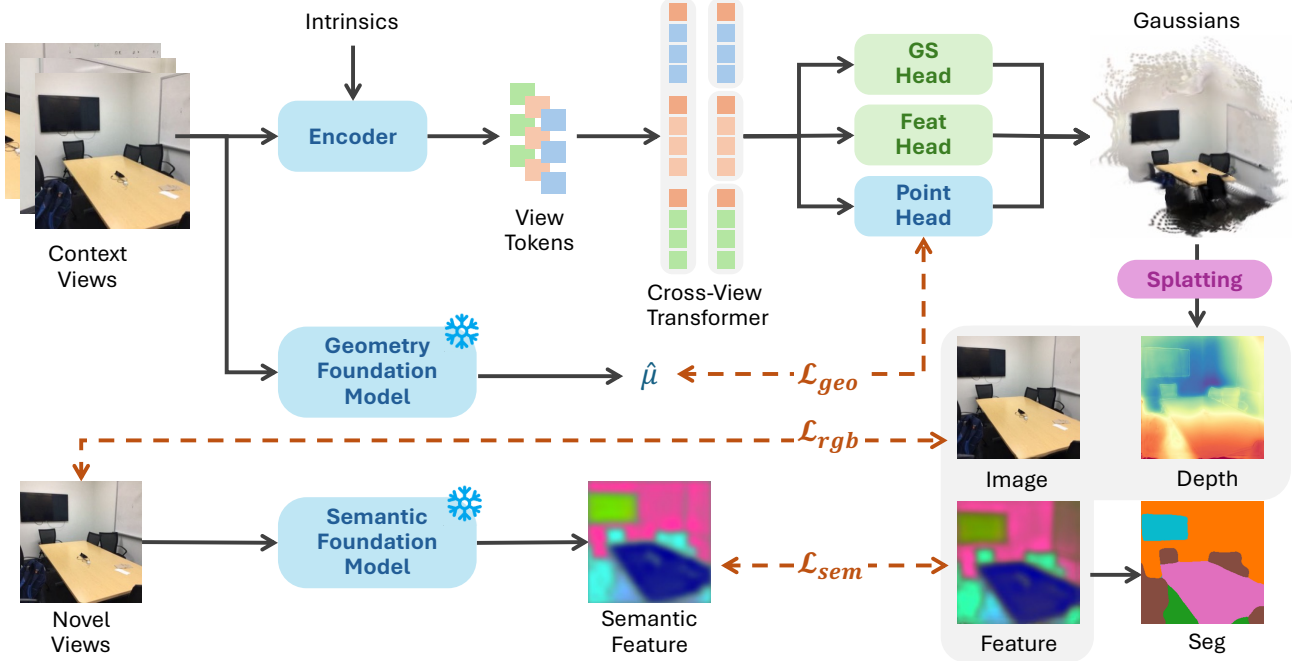


Figure 2. Architectural overview of the Uni3R pipeline. Uni3R predicts a set of Gaussian primitives with jointly integrated geometry, appearance, and open-vocabulary semantics in a single pass, eliminating the need for per-scene optimization.

iLRM [13] and DepthSplat [39] adapt this generalizable paradigm to predict 3D Gaussian parameters directly. However, these approaches typically necessitate known camera poses to guide the reconstruction. To eliminate this constraint, MAST3R [17] and DUST3R [37] demonstrate the feasibility of predicting pixel-aligned 3D point clouds directly from image pairs without explicit pose information. Building on these advances, Splatt3R [32] and NoPoSplat [40] further advance this pose-free paradigm by predicting 3D Gaussian primitives directly from image pairs. Despite their progress, models based on the DUST3R architecture still require sufficient overlap between image pairs and struggle to integrate globally consistent information, leading to fragmented reconstructions. Uni3R overcomes these limitations by employing a Cross-View Transformer, inspired by VGGT [36], to interpret and fuse information from an arbitrary number of views. Based on the globally consistent 3D geometric features, we develop a multi-view, pose-free feed-forward reconstruction model. Our method supports not only image pairs but also extended sequences or video clips, predicting 3D Gaussian primitives in a single forward pass to achieve high-quality, globally coherent 3D reconstruction without requiring camera poses.

2.3. Open-Vocabulary Segmentation in 3DGS

Integrating semantics into 3D reconstructions is crucial for higher-level scene understanding tasks. Early methods for 3D semantic segmentation required dense, 3D ground-truth labels, which are scarce and laborious to acquire. The ad-

vent of powerful 2D vision-language models like CLIP [19, 29] has spurred the development of open-vocabulary methods that lift 2D understanding into 3D. LERF [15] distills 2D CLIP features into 3D radiance fields. Capitalizing on the rendering efficiency of 3D Gaussian Splatting, several methods [21, 28, 43, 45] have extended Gaussian representations with semantic features. Nonetheless, these methods still rely on per-scene optimization, making them unsuitable for real-time applications in novel environments. While generalizable approaches like LSM [8] and GSemSplat [38] have been proposed, they are typically constrained to two-view inputs, restricting their scalability and robustness in complex scenes. In a related vein, GaussTR [11] explores generalizable Gaussian-based segmentation in the context of occupancy prediction. In contrast, Uni3R integrates open-vocabulary understanding into a generalizable, multi-view framework, producing globally consistent 3D representation embedded with expressive semantics without requiring any 3D semantic labels.

3. Method

This section details our methodology, beginning with the Feed-Forward 3D Gaussian Model in Sec. 3.1. We then describe how to endow Gaussians with semantics in Sec. 3.2, and conclude with the specifics of the training losses in Sec. 3.3, including photometric loss, semantic loss and geometry loss.

3.1. Feed-Forward Gaussian Splatting

3.1.1. Intrinsic Embedding

To resolve the inherent scale ambiguity in monocular reconstruction caused by unknown focal lengths, we incorporate an intrinsic embedding to provide essential geometric cues. Following NoPoSplat [40], we encode each camera’s focal length and principal point with a linear projection. The resulting intrinsic embedding is concatenated channel-wise with the corresponding image before patch tokenization, allowing the network to reason about the geometry-aware information.

3.1.2. Cross-View Transformer Encoder

Uni3R employs a Cross-View Transformer Encoder, following VGGT, to extract and fuse features from all input images into a consistent, view-agnostic latent representation. Each input view $I^{(i)}$, augmented with its intrinsic embedding, is first processed by a pre-trained Vision Transformer, DINOv2 [26], to extract a sequence of patch-level feature tokens. To support arbitrary multi-view inputs while maintaining permutation equivariance, a learnable camera token is appended to each view’s token sequence. The Cross-View Transformer Encoder consists of a series of Transformer blocks that alternate between intra-frame and cross-frame attention. Intra-frame self-attention operates within each view’s token set, refining the per-view features with local context. Subsequently, cross-frame global attention aggregates tokens from all views to establish correspondences and reason about the global 3D geometry. The output latent tokens from the encoder encapsulate a holistic and globally consistent understanding of the 3D scene.

3.1.3. Decoding Gaussian Parameters

The fused latent representations are decoded into a dense set of 3D Gaussian primitives with a Dense Prediction Transformer (DPT) [30] followed by dedicated prediction heads for different Gaussian parameters. DPT progressively refines coarse patch-level features with fine-grained local details from intermediate layers, yielding a dense per-pixel feature map.

Subsequently, we predict the properties of a set of pixel-aligned 3D Gaussians with separate MLP heads. Each primitive is parameterized by:

$$G_j = \{\mu_j, \alpha_j, c_j, s_j, r_j, f_j^{\text{sem}}\}, \quad (1)$$

where $\mu_j \in \mathbb{R}^3$ denotes the 3D center point, $s_j \in \mathbb{R}^3$ is the scale, $r_j \in \mathbb{R}^4$ is the rotation quaternion, $\alpha_j \in [0, 1]$ is the opacity, $c_j \in \mathbb{R}^3$ is the color, and $f_j^{\text{sem}} \in \mathbb{R}^d$ is a high-dimensional semantic feature vector.

The point head is initialized from pre-trained VGGT weights and is further fine-tuned with rendering-based supervision to align with real-world metric scales. Distinct

activation functions are applied to the predicted parameters to constrain them to their valid ranges:

$$\alpha_j = \sigma(f_j^\alpha), \quad (2)$$

$$s_j = \exp(f_j^s) \cdot d_{\text{median}}, \quad (3)$$

$$r_j = \text{normalize}(f_j^r), \quad (4)$$

where $\sigma(\cdot)$ represents the sigmoid activation function, and f_j^α , f_j^s , and f_j^r are the latents for opacity, scale, and rotation, respectively. The term d_{median} is the median depth value computed from the predicted 3D positions, which helps to normalize the scale.

3.2. Rendering with Open-Vocabulary Semantics

Once predicted, the set of Gaussians is rendered into novel views using the differentiable 3D Gaussian rasterizer, extended with semantic feature fields. The Gaussian function is described by:

$$G_j(x) = e^{-\frac{1}{2}x^\top \Sigma_j^{-1}x}, \quad (5)$$

where the covariance matrix Σ_j is constructed from the scale s_j and rotation r_j . The rendered color $\hat{I}$ and feature $\hat{F}$ at each pixel are computed by alpha-blending the properties of all sorted Gaussians that overlap it, taking $\hat{F}$ as an example:

$$\hat{F} = \sum_i \hat{f}_i^{\text{sem}} \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j), \quad (6)$$

where $\hat{f}_j^{\text{sem}}$ is compressed from f_j^{sem} by an autoencoder to mitigate the high memory cost of rendering high-dimensional semantic features.

$$\hat{f}_j^{\text{sem}} = \mathcal{F}_{\text{enc}}(f_j^{\text{sem}}), \quad (7)$$

$$\hat{F}' = \mathcal{F}_{\text{dec}}(\hat{F}), \quad (8)$$

where $\mathcal{F}_{\text{enc}}$ and $\mathcal{F}_{\text{dec}}$ are the encoder and decoder, respectively. The autoencoder is trained end-to-end to align the rendered features with CLIP-based image features, enabling efficient open-vocabulary semantic reasoning.

During inference, semantic segmentation is performed by computing the cosine similarity between the pixel-wise semantic features and a set of text-derived prototypes. Given a set of text prompts for desired categories (*e.g.*, “wall,” “chair,” “sofa”), CLIP text encoder generates corresponding feature prototypes $f^{\text{txt}} \in \mathbb{R}^{N_C \times C}$, where N_C is the number of categories. The semantic logits S is then computed by cosine similarity:

$$S_p = \text{softmax}(f^{\text{txt}} \cdot \hat{F}'). \quad (9)$$

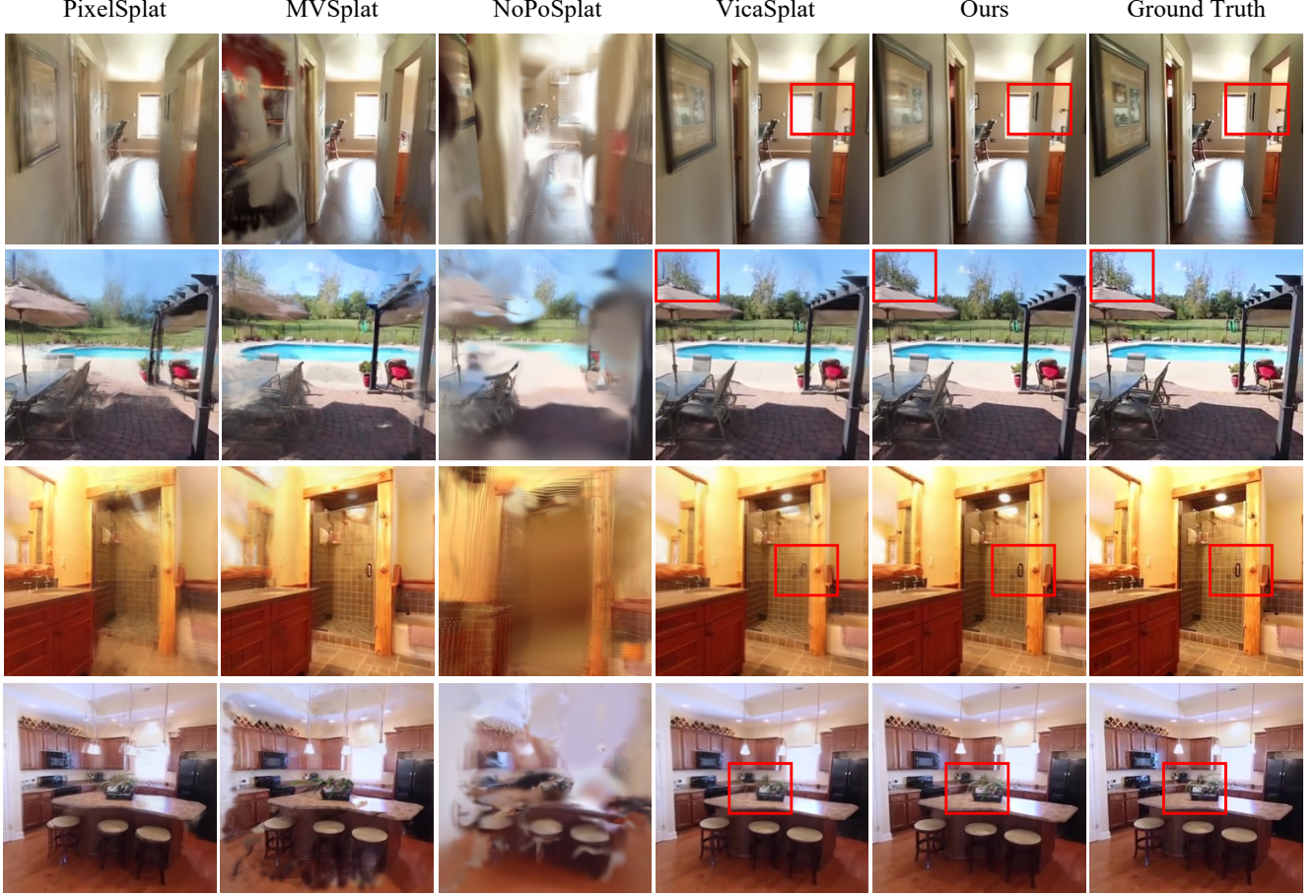


Figure 3. Qualitative comparison of novel view synthesis on RealEstate10k test set with 8 input images.

3.3. Training Objectives

Photometric Loss ($\mathcal{L}_{\text{rgb}}$). To ensure that rendered images match the input views, we combine a pixel-wise L1 loss and the LPIPS metric [44]:

$$\mathcal{L}_{\text{rgb}} = \sum_{i=1}^N \left(\|\tilde{I}^{(i)} - \hat{I}^{(i)}\|_1 + \lambda_{\text{LPIPS}} \text{LPIPS}(\tilde{I}^{(i)}, \hat{I}^{(i)}) \right), \quad (10)$$

where $\tilde{I}^{(i)}$ and $\hat{I}^{(i)}$ denotes the ground-truth image and the rendered image from the i -th camera viewpoint, respectively, and λ_{LPIPS} is set to 0.05.

Semantic Loss ($\mathcal{L}_{\text{sem}}$). To endow the Gaussians with open-vocabulary capabilities, we distill knowledge from a frozen, pre-trained 2D vision-language model, LSeg [18]. We extract feature maps $\tilde{F}^{(i)}$ from each input image using the LSeg image encoder. We then enforce alignment between the rendered semantic feature map $\hat{F}^{(i) \prime}$ and the 2D CLIP-based features using a cosine similarity loss:

$$\mathcal{L}_{\text{sem}} = \sum_{i=1}^N \left(1 - \frac{\tilde{F}^{(i)} \cdot \hat{F}^{(i) \prime}}{\|\tilde{F}^{(i)}\| \cdot \|\hat{F}^{(i) \prime}\|} \right). \quad (11)$$

This loss lifts rich 2D semantics into the 3D domain, enabling zero-shot semantic understanding without requiring explicit 3D annotations.

Geometry Loss ($\mathcal{L}_{\text{geo}}$). To enhance geometric consistency and training stability, we adopt a point-map regularization strategy inspired by PM-Loss [31]. This regularization simultaneously improves structural accuracy—particularly around object boundaries—and mitigates collapse during optimization. Under RGB-only supervision, the model lacks explicit geometric constraints on the predicted point cloud, often leading to local minima and unstable convergence. The introduced point-map constraint provides a strong geometric prior that guides the 3D Gaussian distribution toward both structurally consistent and stable reconstruction. Specifically, we leverage a frozen VGGT [36] to generate a dense point map $\hat{\mu}^{(i)} \in \mathbb{R}^{3 \times H \times W}$ to guide geometric supervision. Given that the predictions from VGGT are not uniformly reliable, especially in challenging regions such as reflective surfaces or areas with heavy occlusion, we introduce a confidence-based masking strategy. We extract the confidence map

Method	Recon. Time↓		Source View					Target View				
	SfM	Per-Scene	mIoU↑	Acc.↑	rel↓	τ ↑		mIoU↑	Acc.↑	PSNR↑	SSIM↑	LPIPS↓
LSeg	N/A	N/A	0.4701	0.7891	-	-		0.4819	0.7927	-	-	-
NeRF-DFF	20.52s	1min	0.4540	0.7173	27.68	9.61		0.4037	0.6755	19.86	0.6650	0.3629
Feature-3DGS	20.52s	18mins	0.4453	0.7276	12.95	21.07		0.4223	0.7174	24.49	0.8132	0.2293
PixelSplat		0.064s	-	-	-	-		-	-	24.89	0.8392	0.1641
LSM*		0.108s	0.5034	0.7740	3.38	67.77		0.5078	0.7686	24.39	0.8072	0.2506
AnySplat		-	-	-	6.35	47.57		-	-	22.08	0.8118	0.2480
Ours		0.162s	0.5403	0.8255	3.87	61.37		0.5584	0.8268	25.53	0.8727	0.1380

Table 1. **Quantitative Comparison on ScanNet.** We evaluate performance on novel view synthesis, depth estimation, and open-vocabulary semantic segmentation. (*) **Unlike LSM, Uni3R is trained without any 3D annotations.**

$C^{(i)}$ from VGGT and construct a binary geometry mask, $M^{(i)} \in \{0, 1\}^{H \times W}$ by selecting the top- k most confident pixels (set as 90% in our experiments). The predicted point maps $\mu^{(i)}$ from Uni3R are then aligned with $\hat{\mu}^{(i)}$ via the Umeyama algorithm [35]. Given the masked aligned point clouds $X_U^{(i)} = \mu^{(i)} \odot M^{(i)}$ and $X_V^{(i)} = \hat{\mu}^{(i)} \odot M^{(i)}$, where $\odot$ denotes the element-wise product, a single-directional Chamfer distance is computed. The loss is formulated as:

$$\mathcal{L}_{\text{geo}} = \sum_{i=1}^N \frac{1}{N_{pts}^{(i)}} \sum_{x \in X_U^{(i)}} \min_{x' \in X_V^{(i)}} \|x - x'\|_2^2 \quad (12)$$

$N_{pts}^{(i)}$ is the total number of points. The final training objective is a weighted sum of the individual losses:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{rgb}} + \lambda_{\text{sem}} \mathcal{L}_{\text{sem}} + \lambda_{\text{geo}} \mathcal{L}_{\text{geo}}, \quad (13)$$

where the balancing hyperparameters λ_{sem} and λ_{geo} are set to 0.02 and 0.005, respectively.

4. Experiments

4.1. Experimental Setup

Dataset For evaluating both 3D scene and semantic field reconstruction, our model is trained on a combined dataset of ScanNet++ [41] and ScanNet [6], totaling 1,565 scenes. We evaluate on 40 unseen ScanNet scenes, and further examine the model’s zero-shot generalization on the Mip-NeRF360 [1] dataset.

Furthermore, to assess rendering quality, we train our model on the RealEstate10K [46] and ACID [22] datasets. To evaluate cross-domain generalization, we test our method on the DTU [10] and ScanNet++ [41] benchmarks (see the supplementary material for more details).

Implementation Details We use DINOv2 [26] as the image encoder, with a patch size of 16, and set the Cross-View Transformer layers as $L = 24$. We initialize the encoder and decoder with the weights from the pretrained

VGGT [36], while the remaining intrinsic layer and Gaussian head are randomly initialized. For a fair comparison with the baseline models, we report all quantitative results under 256×256 . Our model is implemented using PyTorch [27]. All experiments are conducted on $8 \times \text{A100}$ GPUs, taking approximately 22 hours for the training of 2 views, with a batch size of 2. Please refer to the supplementary for more details.

4.2. Experiment Results

Semantic 3D Reconstruction As shown in Tab. 1 and Fig. 4, Uni3R establishes a new state of the art across multiple 3D tasks, producing coherent and precise semantic interpretations. While Uni3R is supervised by LSeg, it outperforms by resolving 2D view-dependent ambiguities through 3D spatial fusion. In Fig. 4, LSeg’s 2D predictions are incorrect for the sofa due to local view. Uni3R, however, aggregates features across multiple views into a unified 3D representation. The underlying multi-view geometry acts as a spatial filter that ‘votes out’ inconsistent 2D errors. Thus, Uni3R not only mimics LSeg, but also leverages 3D consistency to produce a denoised, robust semantic prediction. Furthermore, while methods such as LSM require ground-truth point clouds for supervision, Uni3R eliminates this dependency, demonstrating superior practicality and scalability for real-world applications.

Comparison with Per-Scene Optimized Methods To evaluate efficiency and generalization, we compare Uni3R with the per-scene optimized Feature 3DGS [45]. Such methods rely on Structure-from-Motion [7] to estimate camera poses, leading to high computational overhead and poor scalability. In contrast, as shown in Tab. 2 and Tab. 5, Uni3R demonstrates strong generalization by reconstructing consistent 3D geometry, rendering and semantics from unposed multiview inputs. Notably, it achieves superior performance in both novel view synthesis and open-vocabulary segmentation, offering a substantial speed advantage over traditional per-scene optimization methods.

Method	Feed-Forward	8 views							16 views						
		Time↓	PSNR↑	SSIM↑	mIoU↑	Acc.↑	rel↓	τ ↑	Time↓	PSNR↑	SSIM↑	mIoU↑	Acc.↑	rel↓	τ ↑
Feature-3DGS	✗	$\approx 40min$	18.17	0.674	0.195	0.724	17.28	13.31	$\approx 60min$	17.09	0.649	0.198	0.672	23.71	10.57
Ours	✓	0.359s	24.71	0.851	0.554	0.807	4.25	57.41	0.636s	23.32	0.816	0.558	0.835	3.97	60.48

Table 2. Comparison with Per-Scene Optimized Methods. Time corresponds to the average reconstruction time per scene.

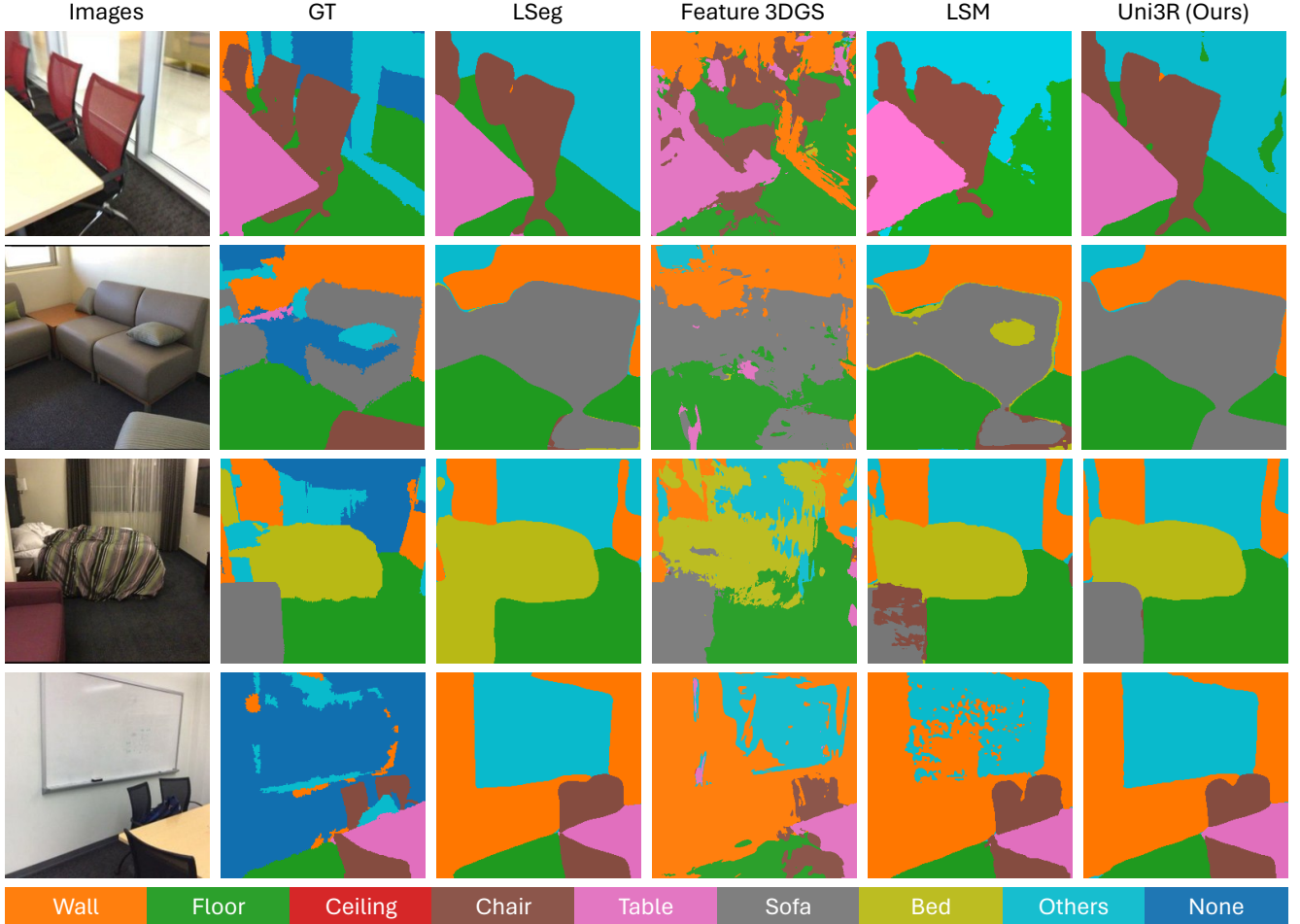


Figure 4. Qualitative Comparison of Novel-View Segmentation on ScanNet.

		RE10k			ACID		
Method		PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
<i>Pose-Required</i>	PixelSplat	23.361	0.803	0.186	25.684	<u>0.778</u>	0.194
	MVSplat	23.430	0.805	0.179	25.335	0.772	0.195
<i>Pose-Free</i>	NoPoSplat	25.036	0.838	0.162	25.961	0.781	0.189
	CoPoNeRF	18.938	0.619	0.388	20.950	0.606	0.406
	VicaSplat	<u>25.038</u>	0.834	<u>0.161</u>	25.439	0.757	0.201
	Ours	25.074	<u>0.837</u>	0.158	<u>25.766</u>	<u>0.778</u>	<u>0.192</u>

Table 3. Quantitative comparisons of novel view synthesis on the RE10k [46] and ACID [22] dataset under 2-views setup.

Novel-View Synthesis As shown in Tab. 3, Uni3R outperforms pose-dependent methods, such as PixelSplat and

MVSplat, by a clear margin (1.7dB), and slightly surpasses baseline model NoPoSplat with a gain of 0.2dB on the RE10k dataset.

Fig. 3 demonstrates Uni3R consistently produces more detailed and structurally coherent constructions. For example, in the Pool Scene, it recovers the forest area with sharper geometry, while VicaSplat shows blurring and discontinuities. These results highlight Uni3R’s ability to preserve fine details and structural consistency in pose-free multi-view 3D reconstruction.

The quantitative results in Tab. 4 further validate the effectiveness of Uni3R in aggregating information across multiple views. Uni3R consistently outperforms all baselines under both 4-view and 8-view settings. Notably, it

	Method	4 views (RE10k)			8 views (RE10k)			4 views (ScanNet)			8 views (ScanNet)		
		PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
<i>Pose-Required</i>	PixelSplat	20.459	0.729	0.267	19.734	0.694	0.290	21.185	0.712	0.351	18.582	0.637	0.440
	MVSplat	20.882	0.761	0.233	19.726	0.743	0.262	14.946	0.472	0.544	13.061	0.407	0.608
<i>Pose-Free</i>	NoPoSplat	16.299	0.552	0.397	14.372	0.469	0.520	16.940	0.560	0.425	12.939	0.397	0.588
	CoPoNeRF	18.299	0.655	0.559	18.984	0.692	0.553	20.247	0.775	0.535	19.821	0.772	0.542
	VicaSplat	24.537	0.814	0.162	24.502	0.806	0.164	26.673	0.856	0.188	23.656	0.777	0.262
	AnySplat	18.860	0.654	-	20.273	0.602	-	21.075	0.722	-	19.303	0.701	-
	Ours	26.360	0.866	0.129	26.629	0.874	0.118	28.324	0.891	0.161	26.019	0.858	0.193

Table 4. Comparison with 4 and 8-view settings on the RE10k [46] and ScanNet [6] datasets.

Method	2View			8View		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Anysplat [12]	17.329	0.401	0.295	18.805	0.434	0.308
Ours	17.331	0.393	0.283	19.196	0.491	0.355

Table 5. Zero-shot generalization on Mip-NeRF360 [1] dataset

View Num	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	mIoU $\uparrow$	Acc $\uparrow$	rel $\downarrow$
2 Views	25.27	0.867	0.141	0.5418	0.8182	4.72
4 Views	25.45	0.845	0.145	0.4018	0.7616	5.76
8 Views	24.76	0.848	0.154	0.5539	0.8184	7.88

Table 6. Arbitrary View model training and evaluation on the ScanNet [6] dataset.

delivers an average improvement of 2.0 dB over VicaSplat [20], a strong sequential baseline designed for unposed video inputs, demonstrating Uni3R’s superior generalization and multi-view integration capabilities. Furthermore, Uni3R surpasses AnySplat [12] on the zero-shot Mip-NeRF360 [1] dataset in Tab. 5 and underscoring its robustness and cross-domain generalization ability.

4.3. Analysis and Ablations

Results on Different Input View Numbers To demonstrate Uni3R’s ability to handle arbitrary view inputs, we report results in Tab. 6. Unlike Vicasplat [20], which focuses solely on sequential rendering, and LSM [8], which reconstructs semantic and radiance fields but is restricted to two-view inputs, Uni3R is the first unified model to jointly reconstruct radiance and semantic fields from unposed multi-view images. This experiment highlights Uni3R’s ability to handle long sequences and wide-baseline configurations, producing high-fidelity and semantically consistent 3D reconstructions in a single feed-forward pass.

Ablation Study on Our Modules We conduct ablation studies on Uni3R to analyze the impact of different supervisory signals and architectural components (see Tab. 7).

Method	mIoU $\uparrow$	rel $\downarrow$	τ $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$
frozen all transformer layers	0.0634	51.1	7.6	5.49	0.052
w/o semantic loss	0.0183	5.8	47.4	25.38	0.869
w/o rendering loss	0.2653	N/A	N/A	N/A	N/A
w/o scale invariant	0.5382	5.8	47.9	24.95	0.861
w/o intrinsic embedding	0.5471	6.5	42.5	25.35	0.871
w/o geo. loss	0.5541	5.6	48.2	25.53	0.872
Ours	0.5484	3.9	61.2	25.53	0.873

Table 7. Ablation Study on different modules. We evaluate the ablated variants of Uni3R, by recording their rendering quality, segmentation performance and geometric accuracy.

Removing the semantic loss causes a severe collapse in segmentation accuracy, underscoring its necessity for open-vocabulary semantic learning. Excluding the rendering loss leads to non-convergence, confirming its critical role in guiding 3D reconstruction. When the geometric loss is removed, the model exhibits degraded 3D consistency (higher depth error and lower τ), validating its effectiveness in improving point cloud distribution and depth alignment. The scale-invariant constraint contributes to rendering stability across scenes with varying depth ranges, while the intrinsic embedding improves robustness by aligning scenes of varying scales into a consistent geometric space. Overall, these results demonstrate that Uni3R’s unified supervision of semantic, radiance, and geometric fields is essential for achieving high-fidelity and semantically consistent 3D reconstruction.

5. Conclusion

Uni3R is a generalizable framework for unified 3D reconstruction and semantic understanding from unposed multi-view images. It predicts a Gaussian-based representation to integrate appearance, geometry, and open-vocabulary semantics in a single forward pass. To address geometric inaccuracies under RGB-only supervision, we introduce a geometry-guided loss to enhance depth consistency. Uni3R takes a significant step toward scalable, multi-view 3D scene understanding for real-world applications, such as autonomous navigation and real-time 3D perception.

Acknowledgements

This research was supported by the Ministry of Science and ICT (MSIT) of Korea, under the National Research Foundation (NRF) grant (RS-2024-00337548) and a grant of the Korea-US Collaborative Research Fund (KUCRF) funded by the Ministry of Science and ICT and Ministry of Health & Welfare, Republic of Korea (grant number: RS-2024-00468417). This work was also supported by Institute for Information & Communications Technology Planning & Evaluation (IITP) grant funded by the Korea government (MSIT) (No. RS-2024-00457882, AI Research Hub Project, No. RS-2025-25441838, Development of a human foundation model for human-centric universal artificial intelligence and training of personnel, and No. RS-2020-II201361, Artificial Intelligence Graduate School Program (Yonsei University).

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